

HCAI5TML01 – Mathematics of Learning.
Theme: Getting Started with Learning Theory.
Lec02 – From Averages to **Guarantees:**
Inequalities in Probability – Part –II.

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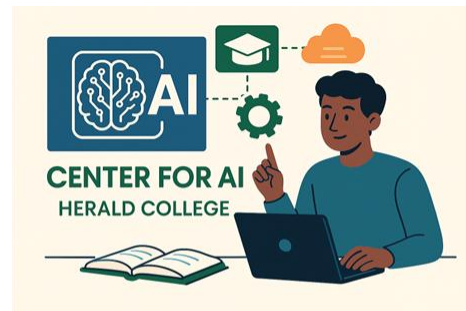


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A. From Last Week ...

A.0 The Intuition behind Bounds.

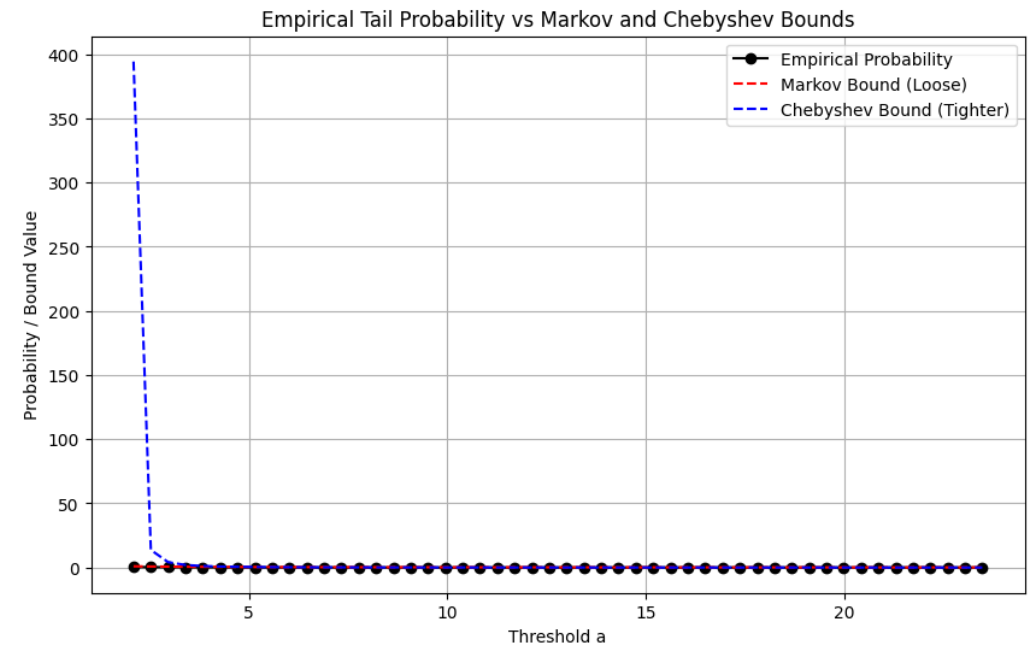
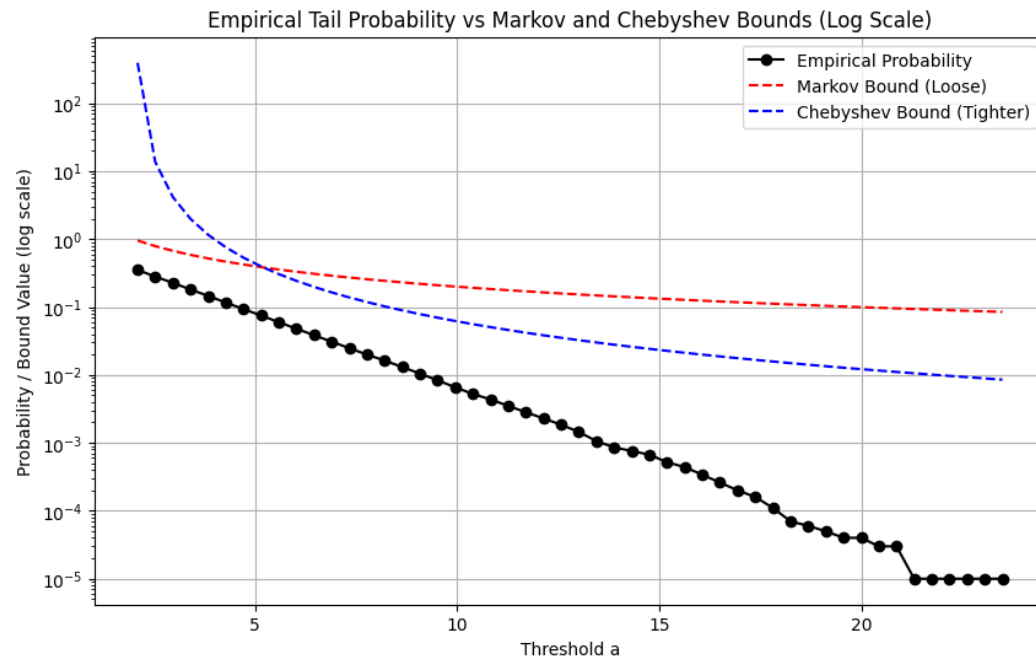
- Defining **Bound**:
 - A **bound** is a **mathematical guarantee** that tells us:
 - “How *large or small a quantity* (like a probability, error, or value) *can be at most (upper bound) or at least (lower bound).*”
 - For example:
 - An **upper bound** on a **quantity Q** means:
 - $Q \leq U$,
 - where **U is the bound**. This says, “*The value of Q will never exceed B.*”
 - A **lower bound** on a **quantity Q** means:
 - $Q \geq L$,
 - where **L is the bound**. This says, “*The value of Q will never be smaller than L.*”
- In the **context of learning theory and probability**:
 - We use **probability bounds** to **guarantee**
 - that the **chance of a rare or “bad” event happening is no more than a certain value.**
 - For example, a **bound on the tail probability** tells us **the probability**:
 - $P(X \geq a) \leq B$ that **X exceeds a is at most B.**

A.0 The Intuition behind Bounds Type.

- Think of **probability bounds** like *closing a door* to keep unlikely events **out**:
 - **Sharp bound**:
 - The door is *exactly* at the right spot:
 - you **block everything** that shouldn't happen and nothing more.
 - **The bound is basically the true probability or asymptotically matches it.**
 - **Tight bound**:
 - The door is *really close* to the right spot
 - maybe you leave a hairline gap, but you're almost matching reality.
 - **For most purposes, it's as good as exact.**
 - **Loose bound**:
 - The door is *way too far* from the real line
 - you **block way too little**, so your bound says *"this event is super likely"* even
 - though the **true probability isn't that large.**

A.0 The Intuition behind Bounds Type.

Say the actual probability is 0.07.		
Bound Type	Value of Bound	Interpretations
Sharp	0.07	Exactly the real probability.
Tight	0.08 or 0.065	Basically, indistinguishable from reality.
Loose	0.5 or 0.3	Wild overestimate – safe but not informative.



A.1 Markov's Inequality – A First Bound on Big Values.

- Idea: : *“If you only know the average, you can still say something about how extreme values behave.”*
- In Simple Terms:
 - If a **non – negative value** (like salary, waiting time or file size) has **known average**, then:
 - The **chance** that it is **very large is small**, unless **the average itself is large**.
- Example – Analogy:
 - Imagine you have a box of apples, and the average weight is 1 kg.
 - Then, *“there can not be many apples heavier than 5 kg”*
 - otherwise, **the average would be higher**.
 - If average = 1kg, then, $\Rightarrow \text{chance of apple} \geq 5 \text{ kg} \leq \frac{1}{5}$

Theorem 1 Markov's Inequality:

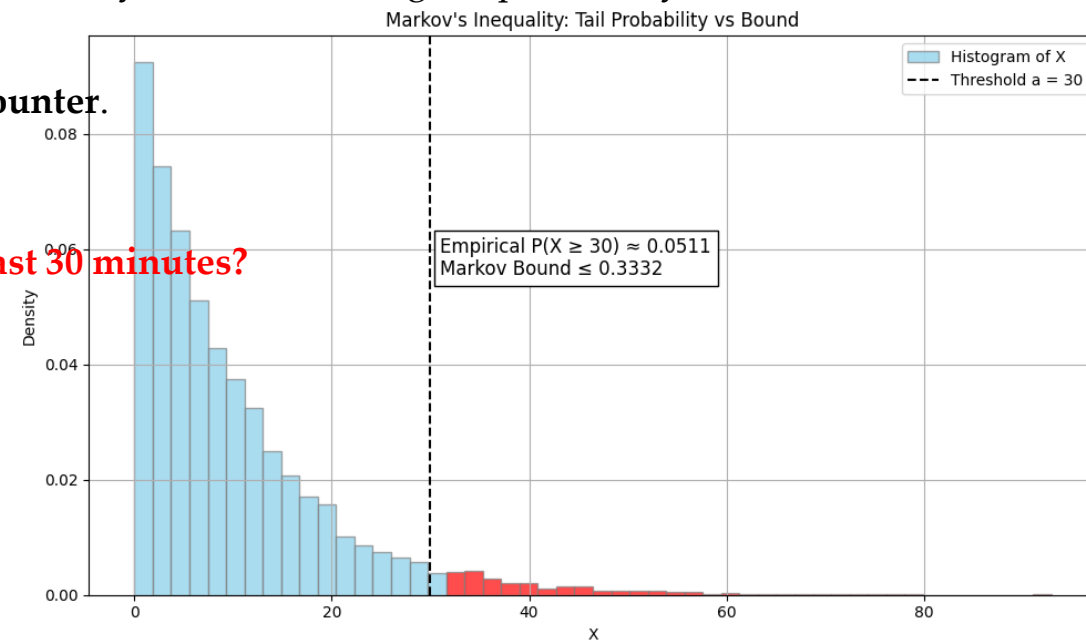
Let $\mathbf{X} \geq 0$ be a non-negative random variable. Then, for any $\varepsilon > 0$, we have

$$\mathbf{P}[\mathbf{X} \geq \varepsilon] \leq \frac{\mathbf{E}[\mathbf{X}]}{\varepsilon}.$$

Markov's Inequality helps bound the **tail probability** of :
a non – negative random variable using only its expectation.

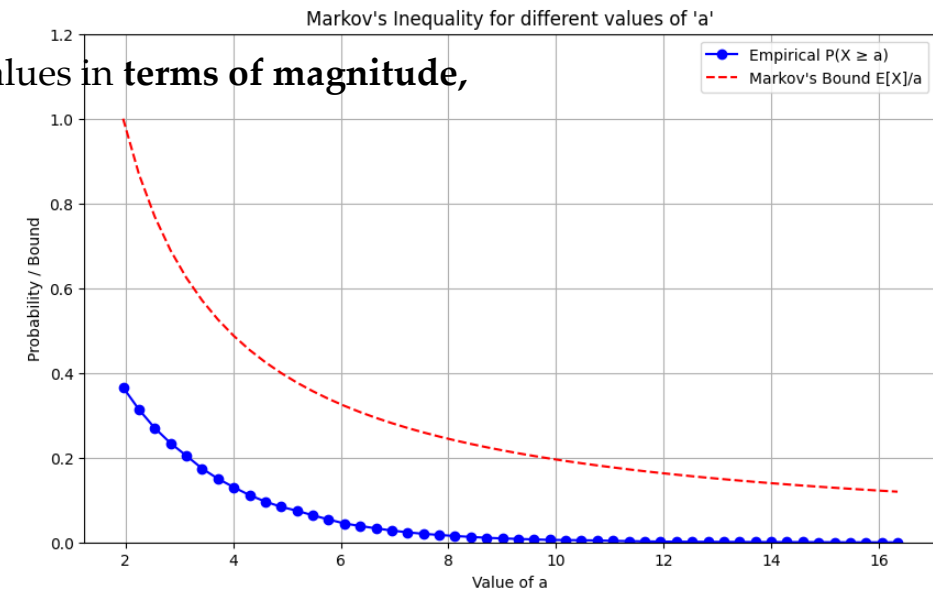
A.1.1 In Markov's Inequality, What does a mean in:

- $\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$
- Interpretation of a :
 - a is a threshold value.
 - It represents a large value of the random variable X , beyond which you are measuring the probability.
 - a can be thought of as the worst-case scenario, for example:
- Suppose X is random variable representing waiting time at a service counter.
 - The average expected withing time is $\mathbb{E}[X] = 10$ minutes
 - You are asking:
 - What's the maximum probability that someone waits at least 30 minutes?
 - So, you set $a = 30$ minutes, and Markov gives:
 - $\mathbb{P}(X \geq 30) \leq \frac{10}{30} = \frac{1}{3}$
 - So, what happens, if you increase the value of a .
 - Markov helps answer:
 - "What is the worst – case chance that X exceeds a ?"



A.1.2 So, what happens, if you increase the value of a .

- If we increase the value of a :
 - The **denominator** in the bound $\frac{\mathbb{E}[X]}{a}$ **increases**,
 - since $\mathbb{E}[X]$ is a **constant** for a **given random variable**,
 - **increasing** the **denominator** will **decrease** the value of the fraction.
 - Therefore, as we **increase** the value of " a " the upper bound of the probability
 - provided by Markov's inequality becomes **smaller (tighter)**.
 - **Cautions:**
 - So, while increasing " a " considers more "extreme" or "worst-case" values in terms of magnitude,
 - it **doesn't make the probability equal to the bound**.
 - It just **provides a tighter upper limit** on that decreasing probability.



A.2 Chebyshev's Inequality – How far can you go from **Average**?

- Idea: “*Most values stay reasonably close to the average – no matter what the distribution looks like.*”
- In Simple Terms:
 - Even if you don't know **the exact shape** of a distribution,
 - but you **know the average and standard deviation**, you can **guarantee** that:
 - Most values lie within a few standard deviations from the mean.*
 - Remember the **Empirical Rule for Normal Distribution**:
 - At least 75% of the values are within 2 standard deviations of the average.
 - At least 89% are within 3 standard deviations.

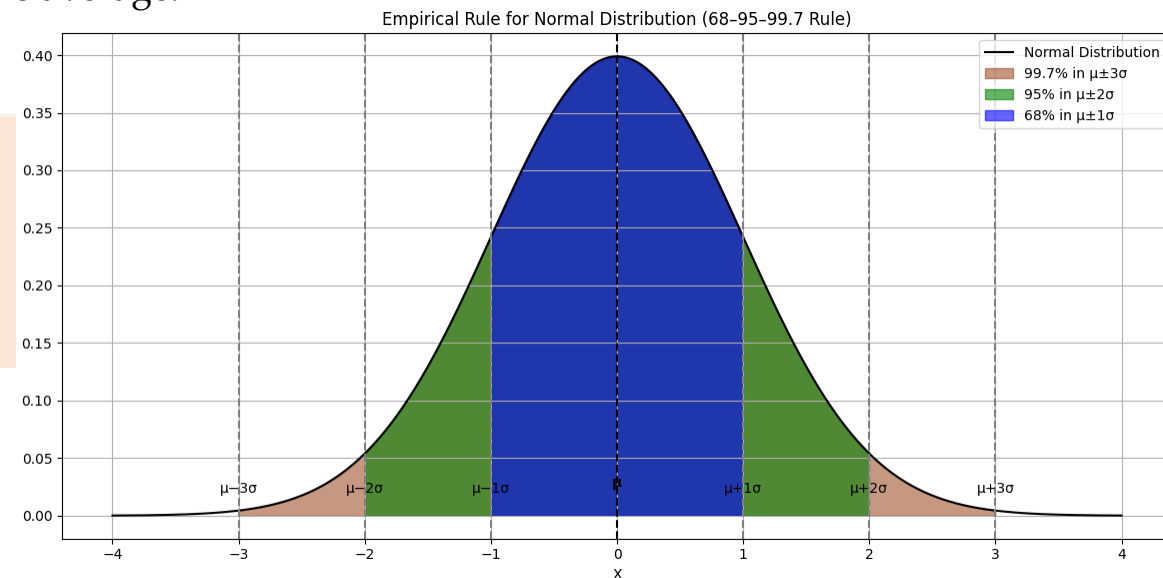
Let **X** be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite non-zero variance $\sigma^2 = \text{Var}(X)$.

Then for any $k > 0$:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \text{ \{ upper Bound \}}$$

or equivalently,

$$\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \text{ \{ lower Bound \}}$$

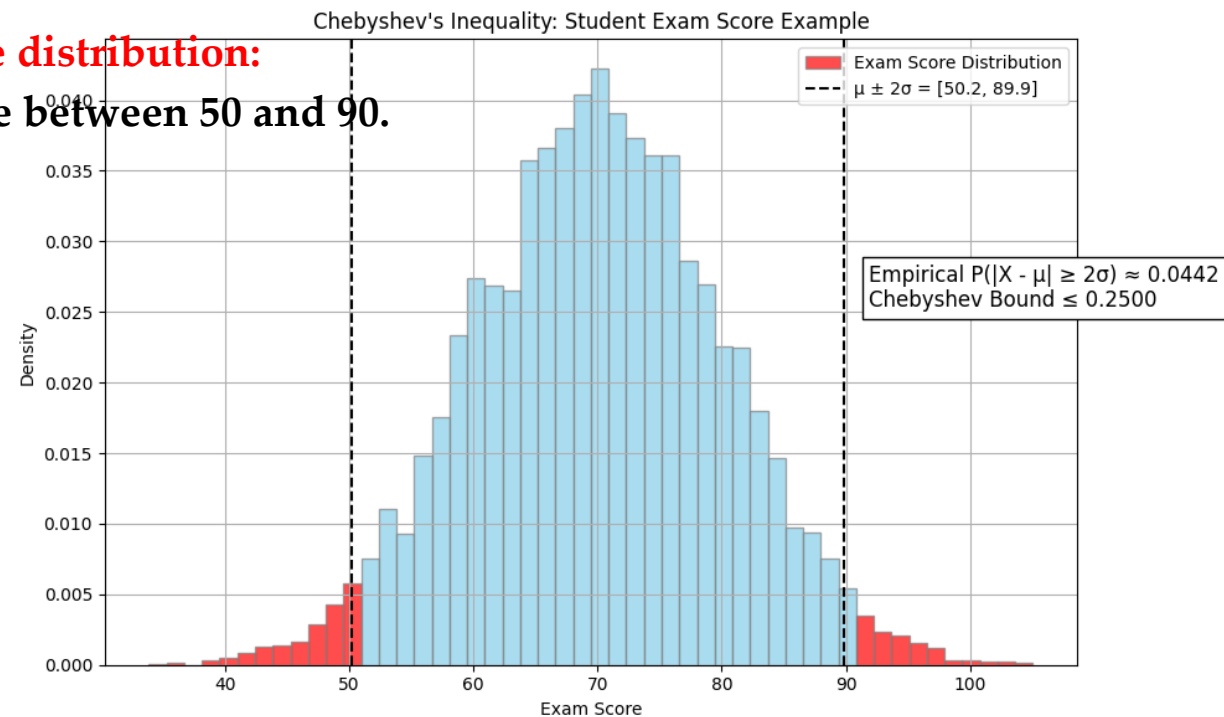


A.2.1 Chebyshev's Inequality Example: Exam Scores in a Classroom

- **Scenario:**
 - You are a teacher analyzing Student performance on a final exam:
 - You calculate:
 - **Mean Score $\mu = 70$**
 - **Standard deviation $\sigma = 10$**
 - You **do not know** the **what the score distribution** looks like
 - it could be normal, skewed, weird but you still want to know:
 - *How many students could have really extreme scores (very far from average)?*
- **Extreme Scores :**
 - What is the **maximum possible probability** that a student scored:
 - **More than 20 ($\Rightarrow 2 \times 10$) marks above or below the mean, i.e. outside the range:**
 - **$\mu \pm 2\sigma = 70 \pm 20 = [50, 90]$**

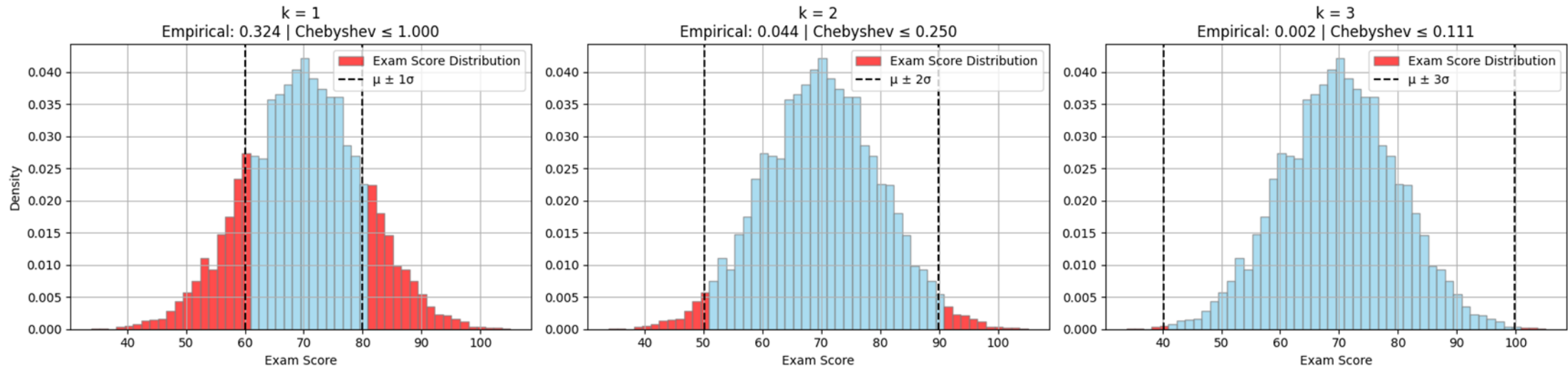
A.2.2 Using Chebyshev's Inequality:

- For $k = 2$, Chebyshev gives:
 - $\mathbb{P}(|X - \mu| \geq 2\sigma) \leq \frac{1}{2^2} = \frac{1}{4} = 0.25$
 - So, **no more than 25%** of students can be **outside the range 50 – 90**.
- Also be interpreted as:
 - Even though we **do not know the shape of the score distribution**:
 - We guarantee that at **least 75% of the scores are between 50 and 90**.



A.2.3 For Different k:

Chebyshev's Inequality for $k = 1, 2, 3$



- Chebyshev's inequality provides an **upper bound** on the probability that a **random variable X** is more **than k standard deviation (σ)** away from its **mean(μ)**:
 - $\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$
- This inequality applies to **any** random variable for which the mean and variance exist.
- From this, we can also infer a **lower bound** on the probability that a value falls **within k standard deviations of the mean**:
 - $\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$
- So, while the **inequality directly bounds** the probability **outside the interval $[\mu - k\sigma, \mu + k\sigma]$**
 - it indirectly tells us that **at least $1 - \frac{1}{k^2}$** of the **probability mass lies within that interval**,
 - for **any random variable with a finite mean and variance**.

3. Chernoff Inequality.

{Summary for Learning Theory.}

3.1 Chernoff Inequality – Intuitive Introduction.

- Setting:
 - You have a **sum** of independent random variables:
 - $\mathbf{X} = \sum_{i=1}^n \mathbf{X}_i$
 - where:
 - Each $\mathbf{X}_i \in \{0, 1\}$ (Bernoulli trials)
 - Let $\mu = \mathbb{E}[\mathbf{X}] = \sum \mathbb{E}[\mathbf{X}_i]$
- Goal:
 - You want to **bound the probability that X deviates from its expected value μ** i.e.
 - *“What is the probability that **the sum** is much larger or smaller than expected?”*
 - What does *“the sum”* mean in this context?

3.1.1 So what is this “*sum*” ?

- We are referring to:
 - $X = \sum_{i=1}^n X_i$
 - where each X_i is random variable – typically a binary variable (e.g. outcome of a coin flip: 0 or 1).
- *Sum* simply refers to the number of successes (or “1”s) observed out of n trials.
 - Example – Coin Flips:
 - Let:
 - $X_i = 1$ if the i^{th} coin flip is heads, and
 - $X_i = 0$ if it's tails.
 - Then: $X = \sum_{i=1}^n X_i \Rightarrow$ is the total number of heads in n flips.
- Now, the expected value of *this sum*:
 - If each coin is fair (i.e. $\mathbb{P}(X_i = 1) = 0.5$), then the expected value is:
 - $\mu = \mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = n \cdot p = n \cdot 0.5$.
 - So, we expect about 500 heads in 1000(n) flips.

3.1.1 So what is this “*sum*” ?

- What Chernoff asks:
 - What's **the probability that X , i.e. the total number of heads,**
 - **is much larger** (say 550) or
 - **much smaller** (say 450) than expected (500)?
 - That's what we mean by:
 - “*What's the probability that the sum deviates significantly from its expectation?*”
 - i.e. **much larger** or **smaller** than **expected**.

3.2 Chernoff Upper and Lower Tail Bound:

- Chernoff Upper Tail Bound:

- For $\delta > 0$:

- $\mathbb{P}(X \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2+\delta}\right)$

- Chernoff Lower Tail Bound:

- For $0 < \delta < 1$:

- $\mathbb{P}(X \leq (1 - \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2}\right)$

3.2.1 What is δ in Chernoff Bounds?

- In **Chernoff's inequality**,
 - δ measures how far you are deviating from the expected value μ – either above or below.
- Specifically:
 - Let's say:
 - You have a **random variable** $X = \sum_{i=1}^n X_i$ (e.g. **number of heads in n coin flips**).
 - The **expected value** $\mu = \mathbb{E}[X]$
 - Then:
 - **Upper tail**:
 - $\mathbb{P}(X \geq (1 + \delta)\mu)$
 - Here, $\delta > 0$ means you are **measuring the chance that X is much larger than expected**.
 - **Lower tail**:
 - $\mathbb{P}(X \leq (1 - \delta)\mu)$
 - Here, $0 < \delta < 1$ means you are **measuring the chance that X is much smaller than expected**.
- **How to compute δ ?**
 - You **calculate** δ based on how far your threshold is from the mean:
 - $\delta = \frac{t - \mu}{\mu}$ for upper tail ($t > \mu$)
 - $\delta = \frac{\mu - t}{\mu}$ for lower tail ($t < \mu$)

3.3 Example Problem: Coin Flips.

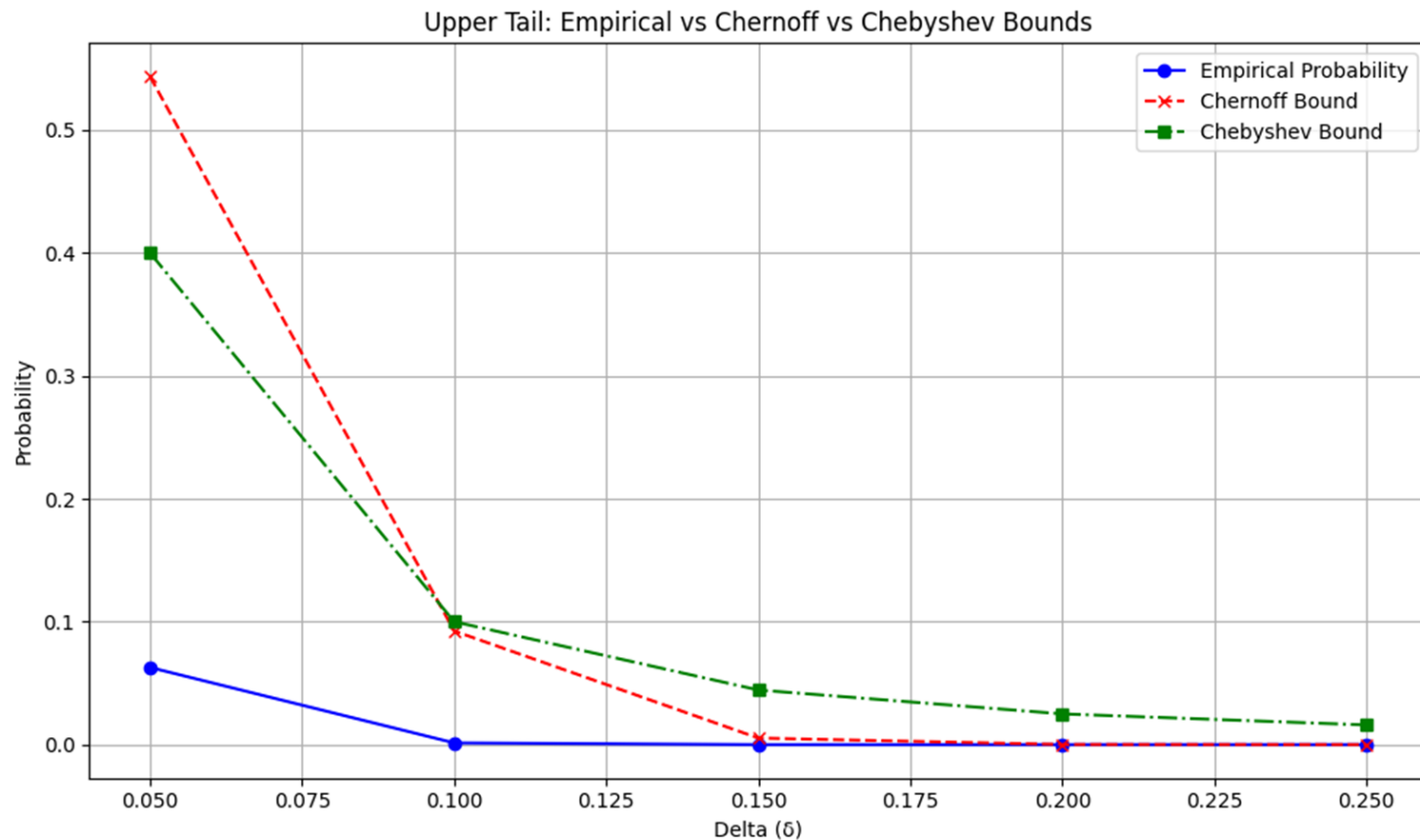
- **Problem Setup:**
 - You are **flipping a fair coin $n = 1000$ times**.
 - Each flip is:
 - A Bernoulli trial (only two outcomes: heads or tails).
 - With probability of heads: **$p = 0.5$** .
 - Let:
 - **X = total number of heads observed**
 - **Expected number of heads:**
 - **$\mu = \mathbb{E}[X] = n \cdot p = 1000 \cdot 0.5 = 500$** .
 - We want to know:
 - **“What is the probability that you get more than 550 heads?”**
 - **$\mathbb{P}(X \leq 550) = ?$**

3.3 Example Problem: Coin Flips.

- Using **Chernoff Inequalities**:
 - Express Deviation as δ :
 - Deviation from the expected value:
 - $\delta = \frac{550 - 500}{500} = 0.1$
 - This means, we are checking the **Probability** that X exceeds the mean by 10% or more.
 - Apply **Chernoff Bound**:
 - The **Chernoff bound** for upper tail is:
 - $\mathbb{P}(X \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2+\delta}\right)$
 - Plug in:
 - $\delta = 0.1; \mu = 500$
 - $\mathbb{P}(X \geq 550) \leq \exp\left(-\frac{(0.1)^2 \cdot 500}{2+0.1}\right) = \exp\left(-\frac{5}{2.1}\right) \approx \exp(-2.38) \approx 0.092$
- So **Chernoff** says:
 - There is **less than a 9.2% chance** you will observe **more than 550 heads** in **1000 flips**.

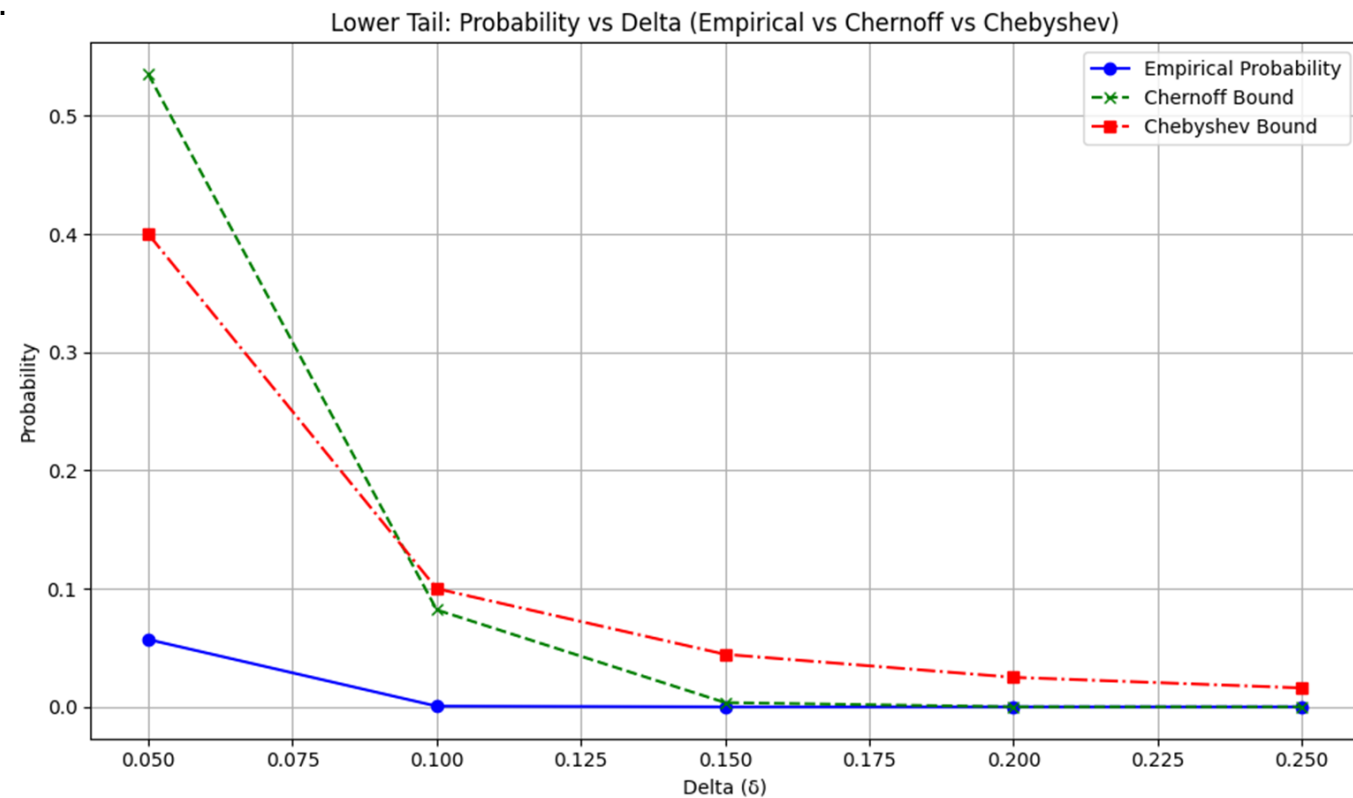
3.4 Visualization: Empirical vs. Chernoff Bound.

- Experiment Parameter:
 - $n = 1000$ # number of trials per experim
 - $p = 0.5$ # probability of success
 - $\mu = n \times p$ # expected value
 - $\sigma^2 = n \times p \times (1 - p)$ #variance
 - trials = 10000 # number of experime
 - $\delta_{\text{values}} = [0.05, 0.1, 0.15, 0.2, 0.25]$



3.4 Visualization: Empirical vs. Chernoff Bound.

- Experiment Parameter:
 - $n = 1000$ # number of trials per experiment.
 - $p = 0.5$ # probability of success
 - $\mu = n \times p$ # expected value
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 - trials = 10000 # number of experiments
 - $\delta_{\text{values}} = [0.05, 0.1, 0.15, 0.2, 0.25]$



4. Chernoff Bound – Theoretical Proof.

{Optional}

4.1 Chernoff Bound – Statement.

- Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be independent random variables taking values in $\{0, 1\}$.

- Let:

- $\mathbf{X} = \sum_{i=1}^n \mathbf{X}_i, \mu = \mathbb{E}[\mathbf{X}] = \sum_{i=1}^n \mathbb{E}[\mathbf{X}_i]$

- Then the Chernoff bound gives exponential tail bounds on

- how far $\mathbf{X}$ can deviate from its expectation.

- Theorem 1 – Upper Tail Bound:

- For $\delta > 0$:

- $\mathbb{P}(\mathbf{X} \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2+\delta}\right)$

- Theorem 2 – Lower Tail Bound:

- For $0 < \delta < 1$:

- $\mathbb{P}(\mathbf{X} \leq (1 - \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2}\right)$

4.2 Chernoff Bound – Proof.

- For upper tail we will prove that:
 - $\mathbb{P}(\mathbf{X} \geq (1 + \delta)) \leq \exp\left(-\frac{\delta^2 \mu}{2 + \delta}\right)$
- Step – 1 – Apply Markov's Inequality:

Applying Markov's Inequality for Chernoff Bound

For any $\lambda > 0$,

$$\mathbf{P}(\mathbf{X} \geq \mathbf{a}) = \mathbf{P}(\mathbf{e}^{\lambda \mathbf{X}} \geq \mathbf{e}^{\lambda \mathbf{a}}) \leq \frac{\mathbf{E}[\mathbf{e}^{\lambda \mathbf{X}}]}{\mathbf{e}^{\lambda \mathbf{a}}}$$

where the inequality follows from Markov's inequality applied to the non-negative random variable $e^{\lambda X}$.

Setting

$$\mathbf{a} = (1 + \delta)\mu$$

where $\mu = \mathbf{E}[\mathbf{X}]$, we obtain:

$$\mathbf{P}(\mathbf{X} \geq (1 + \delta)\mu) \leq \frac{\mathbf{E}[\mathbf{e}^{\lambda \mathbf{X}}]}{\mathbf{e}^{\lambda (1 + \delta)\mu}}$$

Key Insight: This is the foundational step for Chernoff bounds. The parameter $\lambda > 0$ can be optimized later to obtain the tightest possible bound for a given distribution.

Step – 1 – Apply Markov's Inequality:

Step 1: Start with the probability we want, before skip

We want:

$$P(X \geq a)$$

This is the probability that the random variable X takes a value at least as big as a .

Step 2: Exponential trick, before skip

We introduce a parameter $\lambda > 0$ and rewrite the event $X \geq a$ in exponential form:

$$X \geq a \Leftrightarrow e^{\lambda X} \geq e^{\lambda a}$$

So:

$$P(X \geq a) = P(e^{\lambda X} \geq e^{\lambda a})$$

Why do this?

Because exponentials turn sums into products (important for independent variables), and they play very nicely with expectations.

Step 3: Apply Markov's inequality, before skip

Markov's inequality says:

$$P(Y \geq t) \leq \frac{E[Y]}{t}$$

for any non-negative random variable Y and $t > 0$. Here we set:

$$Y = e^{\lambda X}, \quad t = e^{\lambda a}$$

Applying Markov:

$$P(e^{\lambda X} \geq e^{\lambda a}) \leq \frac{E[e^{\lambda X}]}{e^{\lambda a}}$$

Step 4: Simplify, before skip

$$P(X \geq a) \leq e^{-\lambda a} E[e^{\lambda X}]$$

The term $E[e^{\lambda X}]$ is the moment generating function (MGF) of X evaluated at λ :

$$M_X(\lambda) = E[e^{\lambda X}]$$

Step 5: Set a , before skip

Here $\mu = E[X]$ is the mean, and $\delta > 0$ represents the fraction above the mean we care about. So:

$$P(X \geq (1 + \delta)\mu) \leq e^{-\lambda(1+\delta)\mu} M_X(\lambda)$$

Step 6: Minimize over λ

The beauty here: This inequality is true for every $\lambda > 0$. So we choose the λ that minimizes the bound, making it as tight (sharp) as possible.

4.2 Chernoff Bound – Proof.

- For upper tail we will prove that:
 - $\mathbb{P}(\mathbf{X} \geq (1 + \delta)) \leq \exp\left(-\frac{\delta^2 \mu}{2 + \delta}\right)$
- Step – 1 – Apply Markov's Inequality:

Applying Markov's Inequality for Chernoff Bound

For any $\lambda > 0$,

$$\mathbf{P}(\mathbf{X} \geq \mathbf{a}) = \mathbf{P}(\mathbf{e}^{\lambda \mathbf{X}} \geq \mathbf{e}^{\lambda \mathbf{a}}) \leq \frac{\mathbf{E}[\mathbf{e}^{\lambda \mathbf{X}}]}{\mathbf{e}^{\lambda \mathbf{a}}}$$

where the inequality follows from Markov's inequality applied to the non-negative random variable $e^{\lambda X}$.

Setting

$$\mathbf{a} = (1 + \delta)\mu$$

where $\mu = \mathbf{E}[\mathbf{X}]$, we obtain:

$$\mathbf{P}(\mathbf{X} \geq (1 + \delta)\mu) \leq \frac{\mathbf{E}[\mathbf{e}^{\lambda \mathbf{X}}]}{\mathbf{e}^{\lambda (1 + \delta)\mu}}$$

Key Insight: This is the foundational step for Chernoff bounds. The parameter $\lambda > 0$ can be optimized later to obtain the tightest possible bound for a given distribution.

4.2.1 Chernoff Bound – Proof.

- Step – 2 – Use Independence to Factorize:

2. Using Independence to Factorize the Moment Generating Function

Given:

$$X = \sum_{i=1}^n X_i$$

where the X_i 's are independent Bernoulli random variables:

$$E[e^{\lambda X}] = \prod_{i=1}^n E[e^{\lambda X_i}]$$

For each $X_i \in \{0, 1\}$ with $P(X_i = 1) = p_i$:

$$E[e^{\lambda X_i}] = (1 - p_i) \cdot e^{\lambda \cdot 0} + p_i \cdot e^{\lambda \cdot 1} = 1 - p_i + p_i e^{\lambda}$$

Thus, the moment generating function factors as:

$$E[e^{\lambda X}] = \prod_{i=1}^n (1 - p_i + p_i e^{\lambda}) \triangleq M(\lambda)$$

The bound becomes:

$$P(X \geq (1 + \delta)\mu) \leq \frac{M(\lambda)}{e^{\lambda(1+\delta)\mu}}$$

Key Insight: The **independence assumption** allows factorization into a product of simple terms. For identically distributed X_i , this simplifies further to $(1 - p + pe^{\lambda})^n$.

Step – 2 – Use Independence to Factorize:

Step 1: Starting Point

From Markov's inequality, we already have:

$$P(X \geq (1 + \delta)\mu) \leq e^{-\lambda(1+\delta)\mu} E[e^{\lambda X}]$$

for any $\lambda > 0$.

We know:

$$X = \sum_{i=1}^n X_i$$

where X_i are independent Bernoulli random variables with:

$$P(X_i = 1) = p_i, \quad P(X_i = 0) = 1 - p_i$$

Because exponentials turn sums into products:

$$e^{\lambda X} = e^{\lambda \sum_{i=1}^n X_i} = \prod_{i=1}^n e^{\lambda X_i}$$

Taking expectations and using independence:

$$E[e^{\lambda X}] = \prod_{i=1}^n E[e^{\lambda X_i}]$$

Step 3: Compute MGF of Each X_i

Since X_i takes only values 0 or 1:

$$\begin{aligned} E[e^{\lambda X_i}] &= (1 - p_i) \cdot e^{\lambda \cdot 0} + p_i \cdot e^{\lambda \cdot 1} \\ &= (1 - p_i) \cdot 1 + p_i \cdot e^{\lambda} \\ &= 1 - p_i + p_i e^{\lambda} \end{aligned}$$

Step 4: Combine Results

So:

$$E[e^{\lambda X}] = \prod_{i=1}^n (1 - p_i + p_i e^{\lambda})$$

Let's call:

$$M(\lambda) = \prod_{i=1}^n (1 - p_i + p_i e^{\lambda})$$

This $M(\lambda)$ is the moment generating function of X for Bernoulli trials.

Step 5: Final Bound Form

The bound now becomes:

$$P(X \geq (1 + \delta)\mu) \leq e^{-\lambda(1+\delta)\mu} \cdot M(\lambda)$$

and this holds for any $\lambda > 0$. We will later choose λ to minimize this expression to get the tightest bound.

Why This Step Matters

This is where we exploit independence to make the bound computable. Without independence, $E[e^{\lambda X}]$ wouldn't factor nicely. This factorization is what makes Chernoff bounds so much sharper than Chebyshev — we're using more information about the distribution.

4.2.1 Chernoff Bound – Proof.

- Step – 2 – Use Independence to Factorize:

2. Using Independence to Factorize the Moment Generating Function

Given:

$$X = \sum_{i=1}^n X_i$$

where the X_i 's are independent Bernoulli random variables:

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For each $X_i \in \{0, 1\}$ with $P(X_i = 1) = p_i$:

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Thus, the moment generating function factors as:

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Key Insight: The **independence assumption** allows factorization into a product of simple terms. For identically distributed X_i , this simplifies further to $(1 - p + pe^{\lambda})^n$.

4.2.2 Chernoff Bound – Proof.

- **Step – 3 – Optimize the Bound:**

To obtain the tightest possible bound, we minimize the right-hand side with respect to λ :

$$\min_{\lambda > 0} \frac{M(\lambda)}{e^{\lambda(1+\delta)\mu}}$$

Standard calculus optimization (taking derivatives and solving for λ) yields the final form:

$$P(X \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2 + \delta}\right)$$

Key Features:

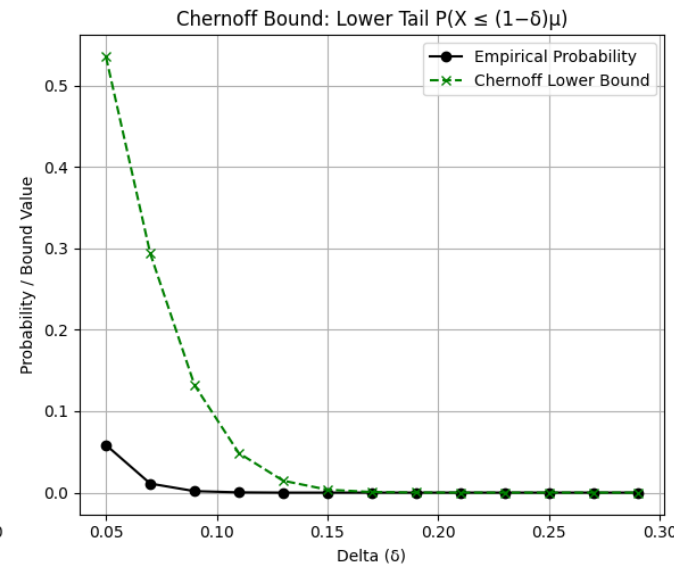
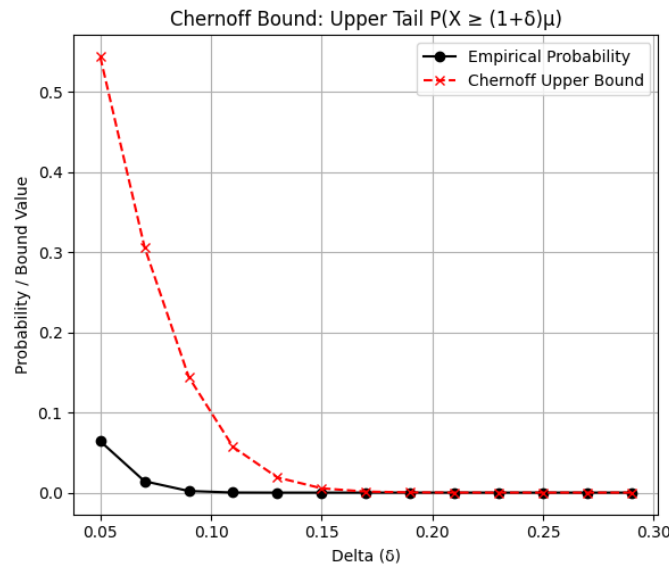
- The bound decays exponentially with δ^2
- For small δ , behaves like $e^{-\delta^2\mu/2}$
- For large δ , becomes $e^{-\delta\mu}$
- Universally valid for any sum of independent Bernoulli trials

This completes the derivation of the upper tail Chernoff bound for sums of independent random variables.

4.2.3 Chernoff Bound – Proof.

- Homework:
 - For Lower bound Tail:
 - A similar derivation using:
 - $\mathbb{P}(X \leq (1 - \delta)\mu) = \mathbb{P}(-X \geq -(1 - \delta)\mu)$ yields:
 - $\mathbb{P}(X \leq (1 - \delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{2}\right)$

Chernoff Bounds for Binomial Distribution (n=1000, p=0.5)



5. Hoeffding's Inequality.

5.1 Hoeffding's In – Equality: The Idea.

- Hoeffding's inequality gives a **bound on the probability**:
 - that the **sum of independent bounded random variables deviates** from **its expected value**.
- It's **particularly** useful when:
 - You know **each variable's min and max** values (boundedness)
 - The variables are **independent** but not necessarily identically distributed

Hoeffding's Inequality - Statement:

Let $X_1, X_2, \dots, X_n$ be independent random variables such that:

$$a_i \leq X_i \leq b_i \quad \text{for all } i = 1, \dots, n$$

Then for any $t > 0$:

$$P\left(\frac{1}{n} \sum_{i=1}^n X_i - E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \geq t\right) \leq \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

where:

- t : Deviation threshold from the mean
- $b_i - a_i$: Range of each random variable
- n : Number of samples
- The bound decays **exponentially** with t^2

Interpretation:

- The inequality bounds the probability that the sample mean deviates from its expected value
- The bound decays exponentially with t^2
- Tighter when the ranges $(b_i - a_i)$ are small
- Applies to any bounded random variables (not necessarily identically distributed)

5.2 What are $\mathbf{a_i}$ and $\mathbf{b_i}$?

- In Hoeffding's Inequality,
 - the values $\mathbf{a_i}$ and $\mathbf{b_i}$ represent the **lower and upper bounds**, respectively,
 - on the possible values of the random variable $\mathbf{X_i}$.
 - where:
 - $\mathbf{a_i}$ = Minimum possible value of $\mathbf{X_i}$
 - $\mathbf{b_i}$ = Maximum possible value of $\mathbf{X_i}$
- Where do we get $\mathbf{a_i}$ and $\mathbf{b_i}$?
 - These **bounds** are derived from the problem setup or known properties of the random variables:
 - **Bounded distributions:**
 - If $\mathbf{X_i}$ is known to lie in a **fixed range** (e.g. $\mathbf{X_i} \in [0, 1]$), then:
 - $\mathbf{a_i} = 0$ & $\mathbf{b_i} = 1$
 - **Physical constraints:**
 - If $\mathbf{X_i}$ represents measurement (e.g. temperature, test scores), the bounds come from **domain knowledge**:
 - E.g. If $\mathbf{X_i}$ is **human height (in cm)** $\Rightarrow \mathbf{a_i} = 50, \mathbf{b_i} = 250$
 - **Theoretical guarantees:**
 - Some algorithms (e.g. **in machine learning**) produce **output with known bounds**.

5.2.1 Why are a_i and b_i important?

1. Tighter bounds:

- The smaller $(b_i - a_i)$, the stronger the inequality.

2. Exponential decay:

- The probability of deviation decreases exponentially with t^2 and inversely with $\sum (b_i - a_i)^2$

- Example:

Given

Let $X_1, \dots, X_n$ be independent random variables where each X_i is bounded:

$$0 \leq X_i \leq 1 \quad (\text{i.e., } a_i = 0, b_i = 1)$$

Hoeffding's Inequality Simplifies to

$$P\left(\frac{1}{n} \sum_{i=1}^n X_i - E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \geq t\right) \leq e^{-2nt^2}$$

Why?

1. Original denominator: $\sum_{i=1}^n (b_i - a_i)^2 = \sum_{i=1}^n (1 - 0)^2 = n$

2. Thus, the bound becomes:

$$\exp\left(-\frac{2n^2t^2}{n}\right) = e^{-2nt^2}$$

Interpretation

- Tighter Bound:** The probability of deviation decays exponentially with n and t^2
- Practical Use:** If $n = 100$ and $t = 0.1$, the bound is $e^{-2 \times 100 \times 0.01} = e^{-2} \approx 0.135$
- Comparison:** For $t = 0.1$, Chebyshev's bound would give $\frac{\text{Var}}{nt^2}$ (much looser for small t)

5.2.1 Why are $\mathbf{a_i}$ and $\mathbf{b_i}$ important?

1. Tighter bounds:

- The smaller $(\mathbf{b_i} - \mathbf{a_i})$, the **stronger the inequality**.

2. Exponential decay:

- The probability of deviation **decreases exponentially with t^2** and **inversely with $\sum(\mathbf{b_i} - \mathbf{a_i})^2$**

• Example:

Why the Simplification Occurs

For variables $X_i \in [0, 1]$:

$$(b_i - a_i) = 1 \quad \text{for all } i \implies \sum_{i=1}^n (b_i - a_i)^2 = n$$

Thus, the general form:

$$\exp\left(-\frac{2n^2t^2}{\sum(b_i - a_i)^2}\right) \text{ reduces to } e^{-2nt^2}$$

Exponential Decay Significance

The bound e^{-2nt^2} has crucial properties:

- Stronger than Chebyshev:** Polynomial decay ($\frac{1}{t^2}$) vs exponential decay (e^{-t^2})
- Preferred for Bounded Variables:** Explains why Hoeffding is widely used in ML for bounded losses

Dependence on Parameters

Parameter	Effect on Bound
Sample size $n \uparrow$	Tighter concentration ($\downarrow$ probability)
Deviation $t \uparrow$	Exponentially smaller probabilities

Implications

- Boundedness is Crucial:** The $[0, 1]$ constraint enables the strong exponential bound
- Practical Impact:** For $n = 1000$, $P(\text{deviation} \geq 0.05) \leq e^{-5} \approx 0.0067$

5.2.2 What is t ?

- In the context of **Hoeffding's Inequality**,
 - the **variable t** represents the **deviation threshold** or **margin of error**
 - for the **sample mean** from its **expected value**.
- **Definition of t :**
 - t is a **positive real number ($t > 0$)** that quantifies:
 - How far the observed sample mean $\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$ is allowed
 - to deviate from the true expected value $\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \right]$.
 - The “*tolerance level*” for the deviation.

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5.3 Example: Hoeffding Inequality in Action.

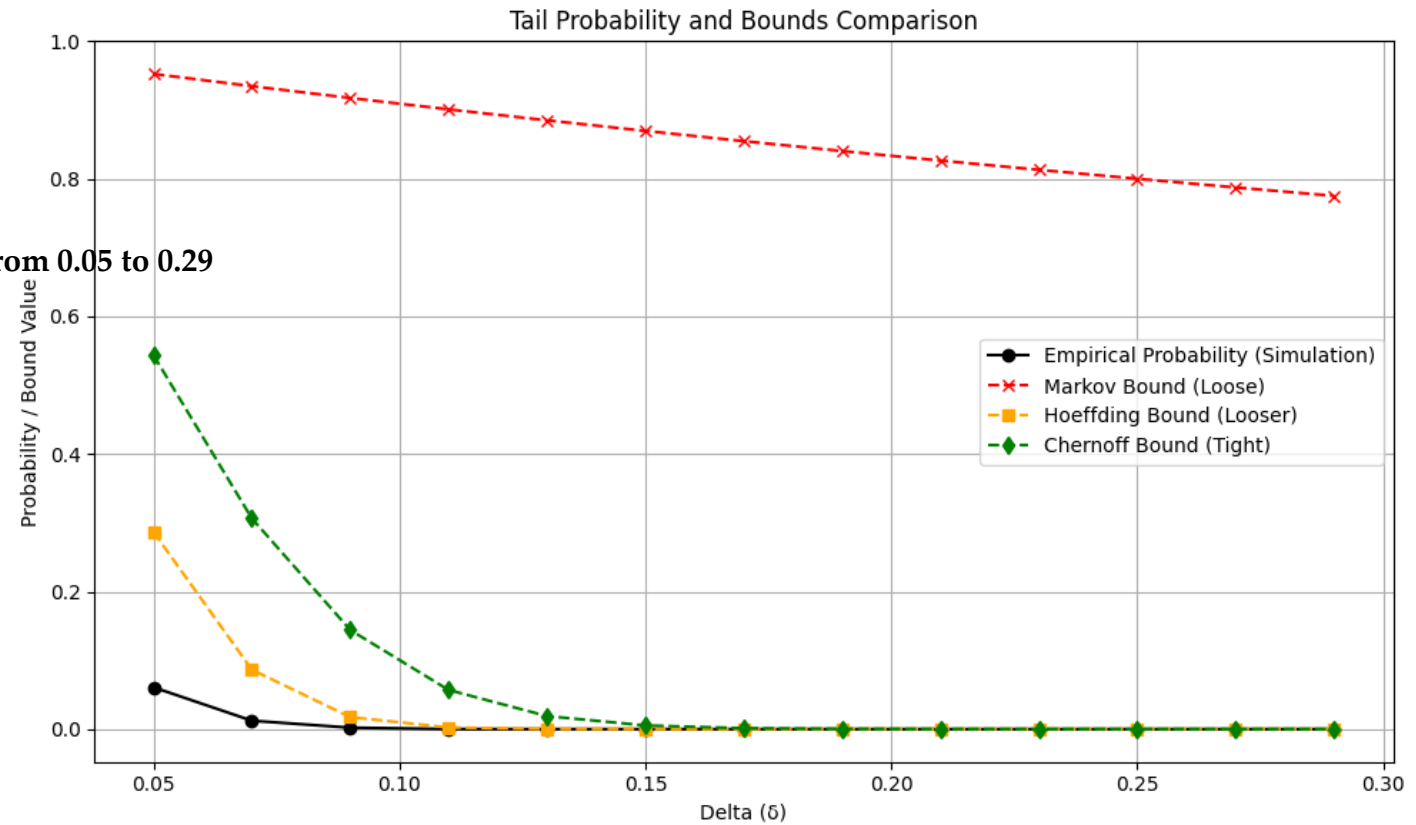
- Scenario:
 - We flip a fair coin $n = 1000$ times.
 - The expected number of heads is:
 - $\mu = n \cdot p = 1000 \cdot 0.5 = 500$
 - We ask:
 - What is the probability of getting at least 550 heads?
 - {That's 50 more heads than expected.}

5.3.1 Example: Hoeffding Inequality in Action.

- Step 1 – Parameters:
 - $t = 550 - 50 = 50$
 - Hoeffding (one-sided bound for independent variables in $[0,1]$):
 - $P(X - \mu \geq t) \leq \exp\left(-\frac{2t^2}{n}\right)$
- Step 2 - Apply Hoeffding:
 - $P(X \geq 550) \leq \exp\left(-\frac{2 \cdot 50^2}{1000}\right)$
 - $P(X \geq 550) \leq \exp\left(-\frac{5000}{1000}\right) = \exp(-5) \approx 0.0067$
 - So Hoeffding tells us:
 - There's less than a 0.67% chance of getting 550 or more heads out of 1000 flips.
- Comparison to Chernoff:
 - Chernoff for same case gave about 9.2% earlier because we used $\delta=0.1$
 - but with its particular form it was looser here.
 - Hoeffding is very sharp for bounded variables like coin flips
 - because it only depends on the range $[0,1]$, not the distribution shape.

5.4 Visualization: Empirical vs Chernoff vs. Hoeffding.

- Experiment parameter:
 - $n = 1000$ # number of trials
 - $p = 0.5$ # success probability
 - $\mu = n * p$ # expected sum
 - trials = 10000 # number of experiments
 - delta_values = [i/100 for i in range(5, 30, 2)] # delta from 0.05 to 0.29



Thank You